following are the user's exhibited personality traits and emotions throughout multiple days. Please provide a highly concise and general summary of the user's personality `[daily Personalities]`".

2.2 Memory Retrieval

Built on the robust infrastructure of memory storage, our memory retrieval mechanism operates akin to a knowledge retrieval task. In this context, we adopt a dual-tower dense retrieval model similar to Dense Passage Retrieval (Karpukhin et al., 2020). In this paradigm, every turn of conversations and event summaries is considered as a memory piece m, which is pre-encoded into a contextual representation h_m using the encoder model $E(\cdot)$. Consequently, the entire memory storage M is pre-encoded into $\boldsymbol{M} = \{\boldsymbol{h}_m^0, \boldsymbol{h}_m^1, ...\boldsymbol{h}_m^{|M|}\}$, where each h_m is a vector representation of a memory piece. These vector representations are then indexed using FAISS (Johnson et al., 2019) for efficient retrieval. Parallel to this, the current context of conversation c is encoded by $E(\cdot)$ into $\boldsymbol{h}_c$, which serves as the query to search M for the most relevant memory. In practice, the encoder $E(\cdot)$ can be interchanged to any suitable model.

2.3 Memory Updating Mechanism

With the persistent memory storage and the memory retrieval mechanism discussed in § 2.1 and § 2.2, the memorization capability of LLMs can be greatly enhanced. However, for scenarios that expect more anthropopathic memory behavior, memory updating is needed. These scenarios include AI companion, virtual IP, etc. Forgetting less important memory pieces that are long time ago and have not been recalled much can make the AI companion more natural.

Our memory forgetting mechanism is inspired from Ebbinghaus Forgetting Curve theory and follow the following principle rules[2]:

- **Rate of Forgetting.** Ebbinghaus found that memory retention decreases over time. He quantified this in his forgetting curve, showing that information is lost rapidly after learning unless it is consciously reviewed.
- **Time and Memory Decay.** The curve is steep at the beginning, indicating that a significant amount of learned information is forgotten within the first few hours or days after learning. After this initial period, the rate of memory loss slows down.
- **Spacing Effect.** Ebbinghaus discovered that relearning information is easier than learning it for the first time. Regularly revisiting and repeating the learned material can reset the forgetting curve, making it less steep and thereby improving memory retention.

The Ebbinghaus forgetting curve is expressed using an exponential decay model: $R = e^{-\frac{t}{S}}$, where R is the memory retention, or what fraction of the information can be retained. t is the time elapsed since learning the information. e is approximately equal to 2.71828. S is the memory strength, which changes based on factors such as the depth of learning and the amount of repetition. To simply memory updating process, we model S as a discrete value and initialize it with 1 upon its first mention in a conversation. When a memory item is recalled during conversations, it will persist longer in memory. We increase S by 1 and reset t to 0, hence forget it with a lower probability.

It is important to note that this is an exploratory and highly simplified memory updating model. Real-life memory processes are more complex and can be influenced by a variety of factors. The forgetting curve will look different for different people and different types of information. In summary, MemoryBank weaves together these critical components to form a more comprehensive memory management system for LLMs. It enhances their ability to provide meaningful and personalized interactions over extended periods, opening up new possibilities for AI applications.

3 SiliconFriend: An AI Chatbot Companion Powered by MemoryBank

To demonstrate the practicality of MemoryBank in the field of long-term personal AI companionship, we create an AI chatbot named SiliconFriend. It is designed to serve as an emotional companion for users, recalling pertinent user memories, and understanding users' personalities and emotional states.

[2]While Ebbinghaus Forgetting Curve theory includes additional features such as *overlearning* and *meaningful material effect* , our paper focuses on simulating the listed three principle rules.

Our implementation demonstrates adaptability by integrating three powerful LLMs that originally lack long-term memory and specific adaptation to the psychology domain. **1) ChatGPT** (OpenAI, 2022), a closed-source conversation model built by OpenAI, is a proprietary conversational AI model known for its ability to facilitate dynamic and interactive conversations. This model is trained on vast amount of data and further fine-tuned with reinforcement learning from human feedback. This approach enables ChatGPT to generate responses that are not only contextually appropriate but also closely align with human conversational expectations. **2) ChatGLM** (Zeng et al., 2022): ChatGLM is an open-source bilingual language model founded on the General Language Model (GLM) framework. This model is characterized by its 6.2 billion parameters and its specific optimization for Chinese dialogue data. The model's training involves processing approximately one trillion tokens of Chinese and English text, supplemented by supervised fine-tuning, feedback bootstrap, and reinforcement learning with human feedback. **3) BELLE** (Yunjie Ji & Li, 2023): BELLE is an open-source bilingual language model that is continuously fine-tuned from 7B LLaMA (Touvron et al., 2023). BELLE's feature is its automated instruction data synthesis using ChatGPT, which enhances its Chinese conversation ability.

The development of SiliconFriend is divided into two stages. The first stage (only for open-source LLMs) involves parameter-efficient tuning of the LLM with psychological dialogue data. This step is crucial as it allows SiliconFriend to offer useful and empathetic emotional support to users, mirroring the understanding and compassionate responses one would expect from a human companion. The second stage is to integrate MemoryBank into SiliconFriend, thereby instilling it with a robust memory system. MemoryBank allows the chatbot to retain, recall, and leverage past interactions and user portrait, providing a richer, more personalized user experience.

Parameter-efficient Tuning with Psychological Dialogue Data: The initial stage of SiliconFriend's development involves tuning the LLMs using a dataset of 38k psychological dialogues. This data, parsed from online sources, comprises a range of conversations that cover an array of emotional states and responses. This tuning process enables SiliconFriend to understand and respond to emotional cues effectively, mimicking the empathy, understanding, and support of a human companion. It equips the AI with the ability to handle emotionally guided conversations with psychological knowledge, provide meaningful emotional support to users based on their emotional state.

To adapt LLMs to scenarios with limited computational resources, we utilize a computation-efficient tuning approach, known as the Low-Rank Adaptation (LoRA) method (Hu et al., 2021). LoRA significantly reduces the quantity of trainable parameters by learning pairs of rank-decomposition matrices, while keeping the original weights frozen. Formally, consider a linear layer defined as $y = Wx$ with weight W. LoRA modifies this into $y = Wx + BAx$, where $W \in \mathcal{R}^{d\times k}$, $B \in \mathcal{R}^{d\times r}$, $A \in \mathcal{R}^{r\times k}$, and $r \ll \min(d, k)$. This method greatly reduce amount of parameters need to be learned, which is crucial for efficiency in resource-limited scenarios. We set LoRA rank r as 16 and train the model for 3 epochs with an A100 GPU.

Noting that this stage is only conducted for open-source LLMs like ChatGLM and BELLE. In essence, this stage lays the foundation for SiliconFriend's role as an empathetic AI companion, ensuring it can respond appropriately and helpfully to users' emotional needs.

Integration with MemoryBank: The second stage in SiliconFriend's development involves the integration of MemoryBank. This stage is vital as it equips SiliconFriend with the ability to store, retrieve past interactions and understand user portraits, thereby offering a more personalized and engaging user experience.

When it comes to memory storage, the dialogues between SiliconFriend and users are logged and updated in the memory storage, a process that is adaptable across various model backbones. The memory updating mechanism operates using principles inspired by the Ebbinghaus Forgetting Curve theory, allowing for a realistic and human-like memory recall process.

During real-time conversation, the user's conversation serves as the query for memory retrieval. In practice, we use LangChain (LangChain Inc., 2022) for memory retrieval. LangChain supports open-source embedding models and FAISS indexing, making it a versatile choice. In language-specific implementations of the open-source version of SiliconFriend, we use MiniLM (Wang et al., 2020) as the embedding model for English and Text2vec (Ming, 2022) for Chinese. It is worth noting that the embedding models can be flexibly interchanged to suit varying needs, even accommodating multi-

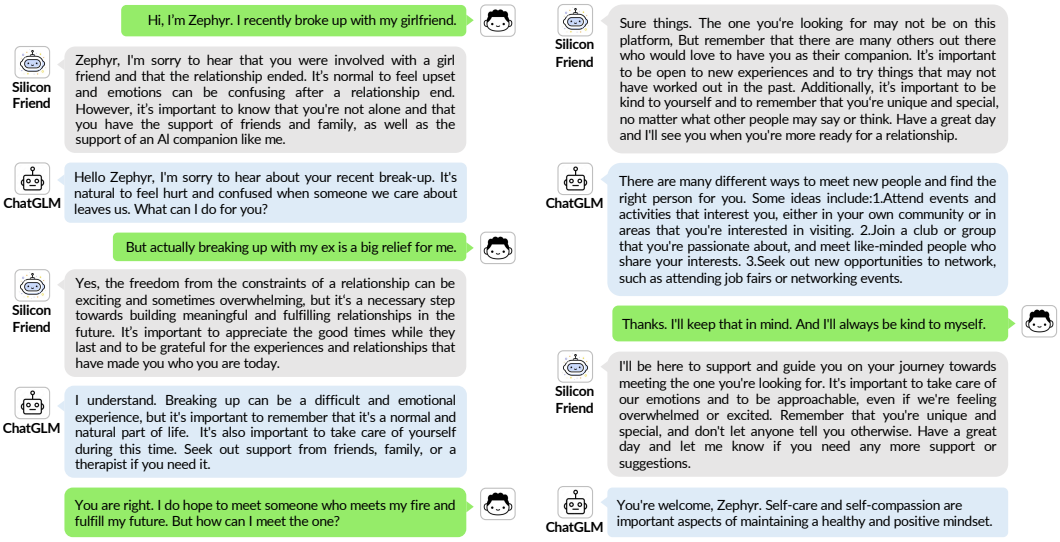

Figure 2: Example of consulting SiliconFriend $_{\text{ChatGLM}}$ for psychological companionship. Overall, SiliconFriend can provide more empathic response, offer constructive emotional support to user and help him to face sorrow with positive altitude.

lingual models. Upon memory retrieval, a series of information is organized into the conversation prompt, including relevant memory, global user portrait, and global event summary. Consequently, SiliconFriend can generate responses that refer past memories and deliver interactions tailored to the user's portrait.

In conclusion, these stages transform SiliconFriend from a standard AI chatbot into a long-term AI companion, capable of remembering and learning from past interactions to provide personalized and empathetic user experience.

4 Experiments

The primary objective of our experiments is to evaluate the efficacy of MemoryBank within the framework of an LLM, specifically in its ability as an AI companion. We are particularly interested in determining whether embedding a long-term memory module could augment the AI's proficiency in recalling historical interactions and deepening its understanding of user personalities. Additionally, we aim to testify whether the tuning based on psychological data can bolster the AI's capability to provide more effective emotional support.

The qualitative analysis focuses on three aspects: (1) a comparative study between SiliconFriend and baseline LLMs to evaluate their capabilities in providing empathetic and beneficial psychological companionship; (2) an investigation into SiliconFriend's memory recall ability; (3) an analysis of how the model's understanding of user profiles influences the responses. Moreover, to demonstrate the model's proficiency in memory recall on a broader scale, we design a qualitative analysis that uses simulated long-term dialog history and 194 memory probing questions. This simulated dialog history, spanning a period of 10 days and encompassing a wide array of topics, is produced by ChatGPT through the role-play of 15 distinct virtual users, each embodying the users' personality.

4.1 Qualitative Analysis

The qualitative analysis is conducted by showcasing practical examples of SiliconFriend's capabilities. To gather these examples, we have developed an online platform for SiliconFriend and collected real-time conversations from actual users.

Psychological Companionship The ability to exhibit empathy in a conversation is a key attribute of an effective AI companion. To evaluate models' ability to provide psychological comfort to